

HAMI LEE

Curriculum Vitae

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EDUCATION

DPhil

Year of completion: Aug. 2023

Institution: University of Oxford

Department: Psychiatry

Supervisors: Dr.Elizabeth Tunbridge and Professor Paul Harrison

Thesis: **Characterization of the short transcript in the C-terminus region of CACNA1S and CACNA1F in human brain**

MSc

Year of completion: Sep 2016

Institution: King's College London

Program: Gene, Environment and Development

Supervisors: Dr. Sylvane Desrivieres

Thesis: **Does DNA methylation affect the white matter integrity in the Internal Capsule and control gene expression? : Epigenome- Wide Association Studies (EWAS) with healthy adolescents**

BA

Year of completion: Dec 2014

Institution: University of Minnesota, Twin Cities

Major: Psychology and Sociology

Thesis: **Saccadic Eye Movements and Oculomotor Inhibitory Control in Attention Deficit/Hyperactivity Disorder Patients**

EMPLOYMENT HISTORY

Postdoctoral Researcher

Time period: April 2026 – Present

Institution: Dongguk University Hospital

Department: NeuroPsychiatry

- Managing four distinct research projects investigating neuronal druggable biomarkers of Autism Spectrum Disorder (ASD) and Bipolar Disorder using patient-derived iPSC neuron models.
- Investigating structural and functional neural connectivity of patient-derived neurons utilizing Multielectrode Array (MEA) systems to capture real-time electrophysiological network dynamics.
- Identifying novel molecular mechanisms, gene expressions, and epigenetic alterations within diseased neurons to discover personalized therapeutic drug targets.
- Managing overall laboratory operations, including optimization of research capacities, procurement of neurobiological reagents, and maintenance of advanced electrophysiological equipment.
- Leading cross-functional laboratory meetings and coordinating complex, multi-centered collaborative operations among clinical teams, external research groups, and technical experts.

Postdoctoral Researcher

Time period: April 2025 – April 2026

Institution: University of Luxembourg

Department: Psychology

- Managing a nationwide project examining epigenetic factors of adverse childhood experiences in relation to mental health disorders
- Archiving and organizing three types of questionnaires collected from nine primary schools, ten residential care homes, and four foster-based care services
- Supervising undergraduate and Master's students responsible for data entry
- Supervising a Master's student in data analysis and thesis writing
- Supervising an intern on a project aligned with the main study
- Developing a test battery to examine executive functions in children
- Supervising seven research assistants administering the test battery during data collection

Postdoctoral Researcher

Time Period: Mar 2024 – Mar 2025

Institution: Temple University

Department: Center for substance abuse research

PI: **Dr. Anjali Rajadhyaksha**

- Designed a novel study using iPSC-derived neurons and brain organoids to examine epigenetic marks associated with substance use
- Integrated advanced statistical analyses with molecular biology techniques to deepen understanding of substance use effects
- Presented project findings at international conferences
- Supervised a lab manager in developing basic wet-lab skills
- Supervised undergraduate students conducting literature reviews

Postdoctoral Research Associate

Time Period: Mar 2023 – Mar 2024

Institution: University of Exeter

Department: Psychology

PI: **Dr. Doretta Caramaschi**

- Coordinated and led two Patient and Public Involvement (PPI) meetings with 13 parents of children diagnosed with ADHD
- Presented research studies to public stakeholders
- Analyzed large-scale longitudinal data using innovative approaches, including unsupervised clustering methods, EWAS, and meta-analyses, to identify DNA methylation biomarkers of co-occurring ADHD and anxiety behaviours
- Awarded HLS lab visiting funding from the University of Exeter to establish an international collaboration
- Designed a new collaborative study
- Presented study results as a poster at a scientific conference and as an oral presentation to researchers at the University of Exeter

SKILLS

Programming Language

- Python, Bash, R, Eprime-3

Data Set

- GTEx (Exon expression data), Zoonomia (PhyloP score), FANTOM5 (CAGE), ALSPAC (DNA methylation and behavioural data), Transcriptomic (Short-read sequencing data, Long-read sequencing data), Genomic data (ChIP-seq data)

Statistics Software

- SPSS, Plink

Molecular Biology

- Polymerase Chain Reaction (PCR), Long-read nanopore sequencing, 5' Rapid Amplification of cDNA ends (5'RACE), Cell culture (HEK 293T, SHSY5Y, iPSC-derived neurons), Plasmid Cloning, Western blotting, Immunofluorescence, Immunoprecipitation, Transfection, Chromatin Immunoprecipitation (ChIP)

Language

- English - Fluent, Korean - Native

PUBLICATIONS

Reviewed Articles

- Paul J. Harrison, Syed M. Husain, Hami Lee, Alejandro De Los Angeles, Lucy Colbourne, Arne Mould, Nicola A.L. Hall, Wilfried Haerty, Elizabeth M. Tunbridge, "CACNA1C (CaV1.2) and other L-type calcium channels in the pathophysiology and treatment of psychiatric disorders: Advances from functional genomics and pharmacoepidemiology," Neuropharmacology, Volume 220, 2022. IF: 4.6 / Area: Biochemistry, Genetics and Molecular Biology
- Yeojung Koh, Maria Noterman-Soulinthavong, Anusha Bangalore, Uapingena P. Kandjoze, Zea Bud, Kamryn L. Noel, Hami Lee, Kathryn Franke, Coral J. Cintron-Perez, Anjali M. Rajadhyaksha, Eric B. Taylor, Andrew A. Pieper, "Astrocytic abnormalities in brain-specific Cacna1c-deficient mice: Implications for BBB impairment in neuropsychiatric diseases associated with CACNA1C mutations.Channels, Volume19, 2025. IF: 3.2 / Area: Biochemistry and molecular biology

- Caramaschi D, Fernández-Barrés S, Casey E, Cruells A, Czamara D, Sharkawy ME, Elliott HR, Fore R, Gairola R, Gruzieva O, Huels A, Lahti J, Lee H, Magnano San Lio R, Malmberg A, Mansell T, Merid SK, Novakovic B, Ott R, Pelegrí D, Rezwan FI, Rifas-Shiman SL, Weiss A, de Lauzon-Guillain B, Duijts L, Grote V, Holloway JW, Koen N, Relton CL, Stein DJ, Zar HJ, Braun JM, Cecil KM, Hivert MF, Hummel S, Jaddoe VWV, Karachaliou M, Karmaus W, Kogevinas M, Koletzko B, Kull I, Melén E, Oken E, Räikkönen K, Saffery R, Vrijheid M, Wright J, Ylton K, Heude B, Felix JF, Bustamante M. Breastfeeding association with DNA methylation in the pregnancy and childhood epigenetics (PACE) consortium. Clin Epigenetics. 2026 Feb 28;18(1):63.

Book Chapters

- Nicola A.L. Hall, Syed M. Husain, Hami Lee, Elizabeth M. Tunbridge, "Chapter Fourteen - Long read transcript profiling of ion channel splice isoforms." Methods in Enzymology, edited by Daniel L. Minor, Henry M. Colecraft, Academic Press, 2021, Pages 345-364. <https://doi.org/10.1016/bs.mie.2021.02.015>. IF : 1.709 / Area : Biochemistry

Non-Reviewed Articles

- (In preparation) Hami Lee, Eilis Hannon, Paul Yousefi, Matthew Suderman, Krasimira Tsaneva-Atanasova, Doretta Caramaschi, "Epigenomic Analysis of ADHD co-occurring with Anxiety in the Avon Longitudinal Study of Parents and Children"
- (In preparation) Hami Lee, Michael Swingle, Raehee Park, Seonhee Kim, Stephanie Dawes, Ilker Sariyer, Anjali Rajadhyaksha Translating mouse studies to humans: Characterization of a Behavior-Specific Genomic Region in CACNA1C Intron 3
- (In preparation) Hami Lee, Syed M. Husain, Lily Waxler, Anjali Rajadhyaksha SNPs of VGCC genes in psychiatric disorder

CONFERENCES

- BAP (The British Association for psychopharmacology) 2019 – Poster
- ECNP Workshop for Early Career Scientists in Europe 2020 – Poster
- ECNP (European College of Neuropsychopharmacology) Congress 2020 – Poster
- European calcium channel conference 2022 – Poster
- European Korean Conference 2023 - Talk
- Epigenomics of common diseases 2023 – Poster
- Midatlantic Bioinformatics Conference 2024 – Poster
- European Calcium Channel Conference 2025 -Talk

CREATIVE ACTIVITY

Public Talk

Idea Festival (IF) Oxford 2020

- Organized and led two public symposia on mental health
- Mental Health in Oxford – What's happening
- Mental Health in Oxford – How Can You Help

Exhibitions

Science Festival 2025

- Participated in a public science festival to teach children and parents about emotion regulation and executive function
- Conducted interactive activities such as making sensory bottles with children and using game-like tasks to assess executive function

Projects

MindMap Oxford

A digital storytelling project in which users share personal experiences linked to specific locations, describing how those places made them feel and how the environments affected their well-being. By reading others' stories, users can recognize that they are not alone in experiencing such emotions

Another Oxford

An immersive audio project that allows users to experience simulated auditory hallucinations, a common symptom

in several psychiatric conditions. While not reproducing clinical hallucinations exactly, the project aims to approximate the experience to foster empathy and understanding

GRANTS

Internal Funding

Funded

Title: HLS Lab visiting funding

Funder: University of Exeter

Role: Main recipient

Effort: 100Period: Aug 2023

TEACHING AND ADVISING

Course

Veterinary Seminar

Seoul National University

Term: Fall 2025

Role: Guest Lecturer

Description: Delivered a lecture on long-read RNA sequencing and ChIP-seq to graduate students in veterinary sciences

Language Development and Disorders/ Numerical Competencies: Development and Disorders (BSc)

University of Luxembourg

Term: Spring 2025, Fall 2025

Enrollment: 60

Role: Guest Lecturer

Description: Undergraduate course for students training to become primary school teachers, with a focus on attention-deficit/hyperactivity disorder (ADHD) and executive functions in children

Teaching with Special Education Needs (MSc)

University of Luxembourg

Term: Fall 2025

Enrollment: 20

Role: Lecturer

Description: MSc course for students training to become secondary school teachers, with a focus on understanding and supporting learners with special educational needs arising from neurodiversity, including ADHD, autism spectrum conditions, and learning difficulties. The course covered theoretical models of neurodiversity, evidence-based classroom adaptations, differentiation strategies, and multidisciplinary teams to promote inclusive education.